

《数据库系统原理》实验报告（2）					
题目：DML 语言实验					
学号		姓名		日期	2025.04.16
实验环境： Docker-desktop 4.40.0 oceanbase-ce 4.3.5.1					
实验步骤及结果截图： (1)创建实验所需的数据库并使用 <pre>create database exp2; use exp2;</pre> <pre>sh-4.4# obclient -uroot@sys -h127.1 -P2881; Welcome to the OceanBase. Commands end with ; or \g. Your OceanBase connection id is 3221498432 Server version: OceanBase_CE 4.3.5.1 (r101000042025031818-b6d5706eb3d2c5f501c7fa646ddb32f3dc87069) (Built Mar 18 2025 18:13:36) Copyright (c) 2000, 2018, OceanBase and/or its affiliates. All rights reserved. Type 'help;' or '\h' for help. Type '\c' to clear the current input statement. obclient(root@sys)[(none)]> create database exp2; Query OK, 1 row affected (0.042 sec) obclient(root@sys)[(none)]> use exp2; Database changed obclient(root@sys)[exp2]></pre> (2)参照表格内容，在数据库中建立图书借阅管理系统相关数据表 ①表 category <pre>CREATE TABLE category (No INTEGER NOT NULL, Name VARCHAR(50) NOT NULL, PRIMARY KEY (No));</pre> <pre>obclient(root@sys)[exp2]> CREATE TABLE category (-> No INTEGER NOT NULL, -> Name VARCHAR(50) NOT NULL, -> PRIMARY KEY (No) ->); Query OK, 0 rows affected (0.049 sec) obclient(root@sys)[exp2]> desc category; +-----+-----+-----+-----+-----+-----+ Field Type Null Key Default Extra +-----+-----+-----+-----+-----+-----+ No int(11) NO PRI NULL Name varchar(50) NO NULL +-----+-----+-----+-----+-----+-----+ 2 rows in set (0.012 sec)</pre> ②表 books <pre>CREATE TABLE books (No INTEGER NOT NULL, Title VARCHAR(100) NOT NULL, Author VARCHAR(50) NOT NULL, publication_year INTEGER NOT NULL, C_no INTEGER NOT NULL, PRIMARY KEY (No),</pre>					

FOREIGN KEY (C_no) REFERENCES category(No)

);

```
obclient(root@sys)[exp2]> CREATE TABLE books (
-> No INTEGER NOT NULL,
-> Title VARCHAR(100) NOT NULL,
-> Author VARCHAR(50) NOT NULL,
-> publication_year INTEGER NOT NULL,
-> C_no INTEGER NOT NULL,
-> PRIMARY KEY (No),
-> FOREIGN KEY (C_no) REFERENCES category(No)
-> );
Query OK, 0 rows affected (0.081 sec)

obclient(root@sys)[exp2]> desc books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11) | NO | PRI | NULL | |
| Title | varchar(100) | NO | | NULL | |
| Author | varchar(50) | NO | | NULL | |
| publication_year | int(11) | NO | | NULL | |
| C_no  | int(11) | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.011 sec)
```

③表 readers

```
CREATE TABLE readers (
    No INTEGER NOT NULL,
    Name VARCHAR(50) NOT NULL,
    Gender VARCHAR(10) NOT NULL CHECK (Gender IN ('Male', 'Female')),
    Tel VARCHAR(15) NOT NULL,
    PRIMARY KEY (No)
);
```

```
obclient(root@sys)[exp2]> CREATE TABLE readers (
-> No INTEGER NOT NULL,
-> Name VARCHAR(50) NOT NULL,
-> Gender VARCHAR(10) NOT NULL CHECK (Gender IN ('Male', 'Female')),
-> Tel VARCHAR(15) NOT NULL,
-> PRIMARY KEY (No)
-> );
Query OK, 0 rows affected (0.062 sec)

obclient(root@sys)[exp2]> desc readers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11) | NO | PRI | NULL | |
| Name  | varchar(50) | NO | | NULL | |
| Gender | varchar(10) | NO | | NULL | |
| Tel   | varchar(15) | NO | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.010 sec)
```

④表 borrow_records

```
CREATE TABLE borrow_records (
    No INTEGER NOT NULL,
    B_no INTEGER NOT NULL,
    R_no INTEGER NOT NULL,
    Borrow_date DATE NOT NULL,
    Return_date DATE,
    PRIMARY KEY (No),
    FOREIGN KEY (B_no) REFERENCES books(No),
    FOREIGN KEY (R_no) REFERENCES readers(No)
```

);

```
obclient(root@sys)[exp2]> CREATE TABLE borrow_records (
-> No INTEGER NOT NULL,
-> B_no INTEGER NOT NULL,
-> R_no INTEGER NOT NULL,
-> Borrow_date DATE NOT NULL,
-> Return_date DATE,
-> PRIMARY KEY (No),
-> FOREIGN KEY (B_no) REFERENCES books(No),
-> FOREIGN KEY (R_no) REFERENCES readers(No)
-> );
Query OK, 0 rows affected (0.108 sec)
```

```
obclient(root@sys)[exp2]> desc borrow_records;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No         | int(11)   | NO   | PRI | NULL    |       |
| B_no       | int(11)   | NO   |     | NULL    |       |
| R_no       | int(11)   | NO   |     | NULL    |       |
| Borrow_date | date      | NO   |     | NULL    |       |
| Return_date | date      | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.010 sec)
```

(3)在数据表中插入内容

INSERT INTO category (No, Name) VALUES

- (1, '计算机'),
- (2, '文学'),
- (3, '历史'),
- (4, '科学');

INSERT INTO books (No, Title, Author, publication_year, C_no) VALUES

- (1, 'Python 编程实战', '张三', 2021, 1),
- (2, '百年孤独', '加西亚·马尔克斯', 1967, 2),
- (3, '明朝那些事儿', '当年明月', 2006, 3),
- (4, '时间简史', '霍金', 1988, 4),
- (5, 'Java 核心技术', 'K.S.霍斯特曼', 2019, 1);

INSERT INTO readers (No, Name, Gender, Tel) VALUES

- (1, '小明', 'Male', '13800138000'),
- (2, '小红', 'Female', '13900139000'),
- (3, '小刚', 'Male', '13600136000'),
- (4, '小美', 'Female', '13700137000'),
- (5, '小强', 'Male', '13300133000');

INSERT INTO borrow_records (No, B_no, R_no, Borrow_date, Return_date) VALUES

- (1, 1, 1, '2024-10-01', '2024-10-20'),
- (2, 2, 2, '2024-10-05', '2024-10-25'),
- (3, 1, 3, '2024-10-10', '2024-10-30'),
- (4, 3, 4, '2024-10-15', '2024-11-01'),
- (5, 2, 5, '2024-10-20', NULL);

```
obclient(root@sys)[exp2]> INSERT INTO category (No, Name) VALUES
  -> (1, '计算机'),
  -> (2, '文学'),
  -> (3, '历史'),
  -> (4, '科学');
Query OK, 4 rows affected (0.013 sec)
Records: 4 Duplicates: 0 Warnings: 0

obclient(root@sys)[exp2]> INSERT INTO books (No, Title, Author, publication_year, C_no) VALUES
  -> (1, 'Python编程实战', '张三', 2021, 1),
  -> (2, '百年孤独', '加西亚·马尔克斯', 1967, 2),
  -> (3, '明朝那些事儿', '当年明月', 2006, 3),
  -> (4, '时间简史', '霍金', 1988, 4),
  -> (5, 'Java核心技术', 'K.S.霍斯特曼', 2019, 1);
Query OK, 5 rows affected (0.014 sec)
Records: 5 Duplicates: 0 Warnings: 0

obclient(root@sys)[exp2]> INSERT INTO readers (No, Name, Gender, Tel) VALUES
  -> (1, '小明', 'Male', '13800138000'),
  -> (2, '小红', 'Female', '13900139000'),
  -> (3, '小刚', 'Male', '13600136000'),
  -> (4, '小美', 'Female', '13700137000'),
  -> (5, '小强', 'Male', '13300133000');
Query OK, 5 rows affected (0.015 sec)
Records: 5 Duplicates: 0 Warnings: 0

obclient(root@sys)[exp2]> INSERT INTO borrow_records (No, B_no, R_no, Borrow_date, Return_date) VALUES
  -> (1, 1, 1, '2024-10-01', '2024-10-20'),
  -> (2, 2, 2, '2024-10-05', '2024-10-25'),
  -> (3, 1, 3, '2024-10-10', '2024-10-30'),
  -> (4, 3, 4, '2024-10-15', '2024-11-01'),
  -> (5, 2, 5, '2024-10-20', NULL);
Query OK, 5 rows affected (0.031 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

(4)使用比较运算符查询 2020 年以后出版的图书的 Title 和 publication_year

```
SELECT Title, publication_year
FROM books
WHERE publication_year > 2020;
```

```
obclient(root@sys)[exp2]> SELECT Title, publication_year
  -> FROM books
  -> WHERE publication_year > 2020;
+-----+-----+
| Title           | publication_year |
+-----+-----+
| Python编程实战 |          2021   |
+-----+-----+
1 row in set (0.003 sec)
```

(5)查询借阅过编号为 3 的图书的读者的 Name 和 Tel

```
SELECT r.Name, r.Tel
FROM readers r
JOIN borrow_records br ON r.No = br.R_no
WHERE br.B_no = 3;
```

```
obclient(root@sys)[exp2]> SELECT r.Name, r.Tel
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> WHERE br.B_no = 3;
+-----+-----+
| Name  | Tel      |
+-----+-----+
| 小美  | 13700137000 |
+-----+-----+
1 row in set (0.022 sec)
```

(6)查询作者姓名中包含“张”字的图书信息

```
SELECT *
FROM books
WHERE Author LIKE '%张%';
```

```
obclient(root@sys)[exp2]> SELECT *
-> FROM books
-> WHERE Author LIKE '%张%';
+-----+-----+-----+-----+-----+
| No  | Title          | Author | publication_year | C_no |
+-----+-----+-----+-----+-----+
| 1  | Python编程实战 | 张三  | 2021             | 1    |
+-----+-----+-----+-----+-----+
1 row in set (0.002 sec)
```

(7)查询所有男性读者的借阅记录，包括读者姓名、借阅图书 Title、借阅日期和归还日期，结果按借阅日期升序排列。

```
SELECT r.Name, b.Title, br.Borrow_date, br.Return_date
FROM readers r
JOIN borrow_records br ON r.No = br.R_no
JOIN books b ON br.B_no = b.No
WHERE r.Gender = 'Male'
ORDER BY br.Borrow_date ASC;
```

```
obclient(root@sys)[exp2]> SELECT r.Name, b.Title, br.Borrow_date, br.Return_date
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> JOIN books b ON br.B_no = b.No
-> WHERE r.Gender = 'Male'
-> ORDER BY br.Borrow_date ASC;
+-----+-----+-----+-----+
| Name  | Title          | Borrow_date | Return_date |
+-----+-----+-----+-----+
| 小明  | Python编程实战 | 2024-10-01  | 2024-10-20  |
| 小刚  | Python编程实战 | 2024-10-10  | 2024-10-30  |
| 小强  | 百年孤独       | 2024-10-20  | NULL        |
+-----+-----+-----+-----+
3 rows in set (0.017 sec)
```

(8)查询女性读者的总人数

```
SELECT COUNT(*) AS Female_Count
FROM readers
WHERE Gender = 'Female';
```

```
obclient(root@sys)[exp2]> SELECT COUNT(*) AS Female_Count
-> FROM readers
-> WHERE Gender = 'Female';

+-----+
| Female_Count |
+-----+
|          2 |
+-----+
1 row in set (0.001 sec)
```

(9)查询借阅时长超过 15 天的借阅记录，包括读者姓名、图书 Title

```
SELECT r.Name, b.Title
FROM readers r
JOIN borrow_records br ON r.No = br.R_no
JOIN books b ON br.B_no = b.No
WHERE DATEDIFF(br.Return_date, br.Borrow_date) > 15;
```

```
obclient(root@sys)[exp2]> SELECT r.Name, b.Title
-> FROM readers r
-> JOIN borrow_records br ON r.No = br.R_no
-> JOIN books b ON br.B_no = b.No
-> WHERE DATEDIFF(br.Return_date, br.Borrow_date) > 15;

+-----+-----+
| Name | Title |
+-----+-----+
| 小明 | Python编程实战 |
| 小刚 | Python编程实战 |
| 小红 | 百年孤独 |
| 小美 | 明朝那些事儿 |
+-----+-----+
4 rows in set (0.007 sec)
```

(10)向 borrow_records 表中插入一条记录(借阅编号为 6, 图书编号为 5, 读者编号为 5, 借阅日期为'2025-03-24', 归还日期为空)。

```
INSERT INTO borrow_records (No, B_no, R_no, Borrow_date, Return_date)
VALUES (6, 5, 5, '2025-03-24', NULL);
```

```
obclient(root@sys)[exp2]> INSERT INTO borrow_records (No, B_no, R_no, Borrow_date, Return_date)
-> VALUES (6, 5, 5, '2025-03-24', NULL);
Query OK, 1 row affected (0.004 sec)

obclient(root@sys)[exp2]> SELECT * FROM borrow_records;

+-----+-----+-----+-----+-----+
| No | B_no | R_no | Borrow_date | Return_date |
+-----+-----+-----+-----+-----+
| 1 | 1 | 1 | 2024-10-01 | 2024-10-20 |
| 2 | 2 | 2 | 2024-10-05 | 2024-10-25 |
| 3 | 1 | 3 | 2024-10-10 | 2024-10-30 |
| 4 | 3 | 4 | 2024-10-15 | 2024-11-01 |
| 5 | 2 | 5 | 2024-10-20 | NULL |
| 6 | 5 | 5 | 2025-03-24 | NULL |
+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)
```

(11)计算每本图书的平均借阅时长，并输出平均借阅时长超过 15 天的图书的 No

```
SELECT br.B_no
FROM borrow_records br
WHERE br.Return_date IS NOT NULL
GROUP BY br.B_no
```

HAVING AVG(DATEDIFF(br.Return_date, br.Borrow_date)) > 15;

```
obclient(root@sys)[exp2]> SELECT br.B_no
-> FROM borrow_records br
-> WHERE br.Return_date IS NOT NULL
-> GROUP BY br.B_no
-> HAVING AVG(DATEDIFF(br.Return_date, br.Borrow_date)) > 15;
```

```
+-----+
| B_no |
+-----+
|    1 |
|    2 |
|    3 |
+-----+
3 rows in set (0.003 sec)
```

Q2 查询借阅过编号为 1 或者编号为 2 图书的读者 No

```
SELECT DISTINCT R_no
FROM borrow_records
WHERE B_no = 1 OR B_no = 2;
```

```
obclient(root@sys)[exp2]> SELECT DISTINCT R_no
-> FROM borrow_records
-> WHERE B_no = 1 OR B_no = 2;
```

```
+-----+
| R_no |
+-----+
|    1 |
|    2 |
|    3 |
|    5 |
+-----+
4 rows in set (0.002 sec)
```

Q3 查询既借阅过编号为 1 又借阅过编号为 3 图书的读者 No

```
SELECT DISTINCT br1.R_no
FROM borrow_records br1
JOIN borrow_records br2 ON br1.R_no = br2.R_no
WHERE br1.B_no = 1 AND br2.B_no = 3;
```

```
obclient(root@sys)[exp2]> SELECT DISTINCT br1.R_no
-> FROM borrow_records br1
-> JOIN borrow_records br2 ON br1.R_no = br2.R_no
-> WHERE br1.B_no = 1 AND br2.B_no = 3;
Empty set (0.003 sec)
```

Q4 建立一个包含图书名字，作者，出版年份和图书类别的视图（赋予列名为 stitle, sauthor, spublicationyear, categoryname）。【create view】

```
CREATE VIEW book_details AS
SELECT b.Title AS stitle,
       b.Author AS sauthor,
       b.publication_year AS spublicationyear,
       c.Name AS categoryname
FROM books b
```

JOIN category c ON b.C_no = c.No;

```
obclient(root@sys)[exp2]> CREATE VIEW book_details AS
-> SELECT b.Title AS stitle,
->        b.Author AS sauthor,
->        b.publication_year AS spublicationyear,
->        c.Name AS categoryname
-> FROM books b
-> JOIN category c ON b.C_no = c.No;
Query OK, 0 rows affected (0.127 sec)
```

```
obclient(root@sys)[exp2]> SELECT * FROM book_details;
```

stitle	sauthor	spublicationyear	categoryname
Python编程实战	张三	2021	计算机
百年孤独	加西亚·马尔克斯	1967	文学
明朝那些事儿	当年明月	2006	历史
时间简史	霍金	1988	科学
Java核心技术	K.S.霍斯特曼	2019	计算机

5 rows in set (0.015 sec)

出现的问题:

(1)输出表格竖线无法对齐的问题

在查询后输出结果时，发现表中竖线无法对齐，以最后一题为例，在建好 view 后对其进行查询，发现表格竖线非常混乱且无法调整。

(2)数据类型设置为 VARCHAR 时会报错

在新建 table 时，第一次我将某字段的数据类型设为 VARCHAR，但表格创建无法成功，代码显示语法错误。

```
obclient(root@sys)[exp2]> CREATE TABLE category (
->   No INTEGER NOT NULL,
->   Name VARCHAR NOT NULL,
->   PRIMARY KEY (No)
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds
to your OceanBase version for the right syntax to use near 'NOT NULL,
PRIMARY KEY (No)
)' at line 3
[172.17.0.2:2882] [2025-04-15 09:06:52.554398] [YB42AC110002-000632CD0C48B469-0-0]
```

(3)创建 book 表时，提示“外键约束错误”

在新建 table 时，第一次我创建 books 表出现外键约束错误的报错。

解决方案:

(1)输出表格竖线无法对齐的问题的解决

经查询资料，这是因为中文字体宽度的问题，与实验结果正确与否无关，解决方法是输出为 txt 或换用其他可视化展示方式。下面是在 MySQL Workbench 8.0 CE 中的输出：

	stitle	sauthor	spublicationyear	categoryname
▶	Python编程实战	张三	2021	计算机
	Java核心技术	K.S.霍斯特曼	2019	计算机
	百年孤独	加西亚·马尔克斯	1967	文学
	明朝那些事儿	当年明月	2006	历史
	时间简史	霍金	1988	科学

(2)数据类型设置为 VARCHAR 时会报错问题的解决

经查询资料，在 Oceanbase 中新建 table 时若使用 VARCHAR 数据类型必须标清长度，我改为 VARCHAR(50)后重新创建就不会报错了。

(3)创建 book 表时，提示“外键约束错误”问题的解决

外键引用的表（如 category）尚未创建，创建 category 后再创建 books 就不会报错了。